

PROFILE

I build the layer between raw data and the numbers people actually trust: ingestion, models, tests, orchestration. An end-to-end Airbyte → dbt → Dagster pipeline that runs from a clean checkout, tests on three layers, and CI that rebuilds the warehouse on every pull request.

What a screenshot of a chart never shows: **finding a defect in the data and in my own model, and saying so out loud**. Building the subscription dashboard I recomputed cohorts from raw tables and found first a defect in the dataset, then an error in my own retention model. Both are written up in the portfolio; the second is now closed by a test.

No commercial employment yet. That is exactly what an internship is for: not the second half of the knowledge, but real data and a team's feedback.

30 tests across three layers, all green	6 projects, every line public	96 roadmap days with working code
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TECHNICAL STACK

Languages	SQL — window functions, CTEs, QUALIFY, PIVOT, GROUPING SETS, EXPLAIN ANALYZE · Python — pandas, Polars, requests, OOP
Transformation	dbt Core — models, tests, macros, snapshots (SCD Type 2), incremental, packages, semantic layer (MetricFlow)
Warehouses	Snowflake (scripts, no production account), PostgreSQL, DuckDB
Orchestration	Dagster — assets, schedules, sensors, partitions, backfill, asset checks · Airbyte self-hosted
Quality & CI	dbt tests, dbt-expectations, Elementary, data contracts, GitHub Actions, Git, Docker
BI	Metabase, Lightdash, matplotlib / seaborn

PROJECTS

Production Pipeline — ingestion, marts, orchestration, quality

- Built a pipeline reproducible from a clean checkout: ingestion through the Airbyte pattern, dbt models staging → marts, Dagster orchestration with daily partitions and idempotent backfill via delete+insert.
- Covered the data with **30 tests across three layers** — 16 dbt core tests, 10 dbt-expectations, 4 Elementary anomaly tests; dbt build run: **PASS=35, ERROR=0**.
- Verified the test suite **negatively**: a script corrupts the raw table on purpose, asserts the tests go red, then restores the database. Both planted errors were caught; downstream went SKIP=19.
- Set up CI on GitHub Actions: two jobs per pull request, with the warehouse rebuilt from CSV by the same ingestion code used locally — one data path instead of three.
- Ran Elementary 0.25.1 on DuckDB by writing the duckdb__edr_multi_value_in macro the package was missing; without it the tests failed on tuple IN.

Dagster · dbt Core · DuckDB · Elementary · GitHub Actions · Docker — [production_pipeline/](#), [dbt_analytics/](#)

SaaS Subscriptions — subscription metrics and a public dashboard

- Built 9 dbt models: staging, dim_subscribers with an SCD Type 2 snapshot, marts for MRR, churn, cohorts, RFM and LTV, plus 5 MetricFlow metrics.
- Found an error in my own mart**: mart_cohort_retention measured the month number from the subscription's own start date, so every cohort had exactly one row — month 0 at 100 %. A not_null test let it through. I rewrote the model as a survival curve and guarded it with assert_retention_monotonic: retention cannot rise.
- And a defect in the data**: in the 800-subscription teaching dataset the start date is generated independently of signup, so recomputed cohorts produced retention that grows (11 → 32 → 51 → 71). I removed the untrustworthy metric and stated the limitation on the page itself.
- Shipped the public dashboard: filters recompute the charts from rows, every figure expands to the SQL behind it, and beside them sit a lineage graph down to the model and a quality panel with the latest run.

dbt Core · MetricFlow · DuckDB · SQL · inline SVG — [dbt_analytics/models/](#), [docs/](#)

- Wrote an OOP Python ETL (Factory, Repository, Pipeline) and loaded a star schema: `fct_orders` with 5,000 orders and four dimensions.
 - Handled deliberate source defects: 100 duplicate orders, 5 % NULLs in `channel` and `payment`, foreign keys pointing outside the product dimension.
 - Diagnosed my own mistake: 3,748 orders carried a zero price because the merge on a missing product produced NULL and the ETL filled it with zero. The right call is a quarantine table with a count, and that is written up in the README.
- Python (OOP) · DuckDB · dbt · matplotlib — project/

- Designed and wrote a bot on `python-telegram-bot`: schedules for students, absence marking for teachers, substitutions entered as plain text through a parser of my own, notifications to the affected classes and a daily post to the channel.
 - Google Sheets as the source of truth with automatic import; broadcasts batched under Telegram's ~30 messages per second limit. Designed for ~3,000 students and 50–150 teachers.
- Python · python-telegram-bot · SQLite · Google Sheets API — code on request

EDUCATION	ADDITIONAL
Analytics Engineer roadmap, 96 of 180 days — every day backed by a commit with working code.	• Languages: Russian — native; English — reads technical documentation fluently and works from English sources, spoken in progress. • Freelance design: client work, branding and layout, deadlines and revisions.